

Yudong Xie

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EDUCATION

Tsinghua University, Beijing, China

Sep 2022 – Present

B.E. in Electronic Engineering (expected June 2026)

- **GPA: 3.92/4.0 Rank: 10/259**

Cornell University, Ithaca, New York, United States

Jan 2025 – May 2025

Exchange Student in Electrical and Computer Engineering

- **GPA: 4.3/4.3**

RESEARCH INTERESTS

- **Micro-electromechanical Systems (MEMS)**
- **Intelligent Sensors and Wearable Devices**
- **Human-Machine Interaction (HMI)**

EXPERIENCE

Piezoelectric Micromachined Ultrasonic Transducers (PMUTs) |

May 2025 – Sep 2025

Supervisor: Prof. Liwei Lin, University of California, Berkeley

- An Acoustic Touch-Motion Button with Haptic Function via an In-Situ Fabricated Elastomeric Lens atop PMUTs
 - Developed a multifunctional interface based on a PMUTs array, enabling gesture recognition, pressure estimation, and haptic feedback
 - Designed the PCB to enable automatic transmitting-receiving switch
 - Used a signal processing method to analyze the echo, achieving 5% error rate on pressure estimation
- Wearable PMUTs Array for Multi-Target Transcranial Neuromodulation via Ultrasound
 - Designed an underwater sound pressure field measurement system based on a 3D printer, which can automatically scans 3D space in a serpentine manner and records the sound pressure at the current position
 - Helped to build up the drive system for the PMUTs

Active Acoustic Sensing | Supervisor: Prof. Cheng Zhang, Cornell University

Jan 2025 – June 2025

- EchoForce: Continuous Grip Force Estimation from Skin Deformation Using Active Acoustic Sensing on a Wristband
 - Aiming to predict applied grip force from skin deformation using active acoustic sensing (FMCW combined with cross-correlation) on a wristband
 - Proposed different testing metrics, verified and improved the generalization performance of the method
 - Helped with building the hardware system, and built the ground truth acquisition system

Silent Speech Interface | Supervisor: Prof. Tianling Ren, Tsinghua University

Sep 2023 – Nov 2024

- Silent Speech Sentence Recognition with Six-Axis Accelerometers Using Conformer and CTC Algorithm
 - Attempting to design a system using accelerometers to capture the facial movements of a silent speaker and then translate the signals into words and sentences
 - Designed the FPCB, collected data from patients in the hospital and proposed CTC algorithm to decode the data, achieving a mean accuracy of more than 97% in long sentence recognition
- Artificial Throat (National college students' innovation and entrepreneurship training)
 - As team leader for a national-level project with funding of ¥50,000 (~\$7000)
 - The artificial throat is graphene-based intelligent, wearable device for speech recognition and interaction.

PUBLICATIONS

Silent Speech Sentence Recognition with Six-Axis Accelerometers using Conformer and CTC Algorithm

Yudong Xie, Zhifeng Han, Qinfan Xiao, Liwei Liang, Luqi Tao, Tianling Ren

2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026)

<https://ieeexplore.ieee.org/document/11462945>

An Acoustic Touch-Motion Button with Haptic Function via an In-Situ Fabricated Elastomeric Lens atop PMUTs

Declan Fitzgerald*, Yudong Xie*(*co-first author*), Sean Isomatsu, Nikita Lukhanin, Zihan Wang, Liwei Lin

The 39th International Conference on Micro Electro Mechanical Systems (IEEE MEMS 2026)

<https://ieeexplore.ieee.org/document/11419575>

EchoForce: Continuous Grip Force Estimation from Skin Deformation Using Active Acoustic Sensing on a Wristband

Kian Mahmoodi*, Yudong Xie*(*co-first author*), Tan Gemicioglu*, Chi-Jung Lee, Jiwan Kim, Cheng Zhang

In Proceedings of the 2025 ACM International Symposium on Wearable Computers (ISWC 2025)

<https://dl.acm.org/doi/10.1145/3715071.3750405>

AWARDS AND HONORS

2023 National Scholarship (4/261) (~\$1400)

2024 National Scholarship (9/263) (~\$1400)

2025 "December 9th" Scholarship (Top 2% in Tsinghua) (~\$2800)

2024 Tsinghua Alumni Zhihua Integrated Circuit Scholarship (~\$1400)

Meritorious Winner of 2024 Mathematical Contest in Modeling

First Prize in The 39th National Regional College Student Physics Competition (Non-Physics Group A)

Champion of 2023 Ricoh Hackathon (Beijing, Tianjin and Hebei) (~\$4000)

Outstanding Innovation Award in 2023 Ricoh Hackathon Final

Champion of 2024 Beijing College Students Volleyball League (Division B)

Champion of 2024 Beijing College Students Beach Volleyball Championship (Division B)

Champion of 2024 Tsinghua University John Mo Cup (Men's Volleyball)

Gold Award in 2024 Tsinghua University Student Social Practice

LEADERSHIP

Department of Electronic Engineering Men's Volleyball Team | *Captain*

Sept 2023 – Dec 2024

- Organized a volleyball team of 27 members, arranged training, and taught 20 freshmen to play
- Won our team's **first championship** in a decade
- Won both the **volleyball and beach volleyball championships** in the same year.

SKILLS

Language Proficiency: English (TOFEL: 107/120), Chinese (Native)

Coding Languages: Python, Matlab, C/C++

Technologies: Arduino, $\LaTeX$, Origin, ESP-IDF